Quantifying Retriever–Generator Alignment in RAG with Local Explanations*

Extended Abstract

Korbinian Randl[1,*], Guido Rocchietti[2,*], Aron Henriksson[1], Ziawasch Abedjan[2], Tony Lindgren[1] and John Pavlopoulos[1,3,4]

[1]Department of Computer and Systems Sciences, Stockholm University, Sweden
[2]BIFOLD, Technische Universität Berlin, Germany
[3]Department of Informatics, Athens University of Economics and Business, Greece
[4]Archimedes, Athena Research Centre, Greece

Abstract

Retrieval-Augmented Generation (RAG) couples a dense retriever with a language model, yet the interaction between the two components is opaque. We summarise RAG-E, an end-to-end explainability framework that attributes the retriever's scores to input tokens via Integrated Gradients (with an empirically selected `[unk]` baseline) and the generator's output tokens to context documents via a Monte Carlo-stabilised Shapley approximation (PMCshap). On top of these attributions we define WARG, the Weighted Alignment between Retriever and Generator, which quantifies how closely the generator's document usage follows the retriever's ranking. Experiments with two dense retrievers and three open-weight generators on PopQA, QAMPARI and TREC CAsT reveal substantial misalignment: in up to 77% of the queries the top-ranked document is largely ignored, and in up to 83% the generator relies mostly on a document ranked outside the top 3. WARG separates agreement from disagreement more sharply than Spearman and Pearson correlation, and high WARG is significantly associated with correct answers.

Keywords

Retrieval-Augmented Generation, Explainability, Feature Attribution, Shapley Values, Integrated Gradients, Retriever–Generator Alignment

1. Introduction

Retrieval-Augmented Generation [2] (RAG) is now the default architecture for open-domain question answering, from everyday assistants to high-stakes domains such as medicine and law [3, 4]. Grounding answers in retrieved documents reduces the opacity of large language models (LLMs), but it does not make the pipeline transparent: model-generated citations are statistical in nature and not necessarily faithful [5]. Existing work explains either the retriever (Ret) in isolation [6, 7] or the generator (Gen) in isolation [8, 9], and work on the Ret–Gen "preference gap" [10, 11, 12] adapts one component to the other without measuring how information actually flows between them. This leaves a simple question unanswered, **RQ:** *How can the agreement between Ret and Gen be quantified?* Our EMNLP 2026 paper [1] answers it with two contributions (Figure 1). **(i) Attribution methods tailored to RAG.** We adapt Integrated Gradients [13] (IG) to dense retrievers by systematically selecting a baseline embedding, and introduce PMCshap, a Monte Carlo-stabilised variant of Kernel-SHAP [14] (Kshap) yielding more accurate and reproducible Shapley Value [15] (SV) approximations for autoregressive generators. **(ii) A metric for Ret–Gen alignment.** We propose and validate WARG, which quantifies

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*This extended abstract presents already published results: the full paper [1] is available as a preprint at https://arxiv.org/abs/2601.21803 and will appear in the Proceedings of EMNLP 2026. The Python package is available on PyPI (`pip install rag-exp`); the source code is at https://github.com/k-randl/Interpretable_RAG.

*Corresponding authors.
korbinian.randl@dsv.su.se (K. Randl); guido.rocchietti@tu-berlin.de (G. Rocchietti)
0000-0002-7938-2747 (K. Randl); 0009-0004-9704-0662 (G. Rocchietti); 0000-0001-9731-1048 (A. Henriksson); 0000-0002-2846-1373 (Z. Abedjan); 0000-0001-7713-1381 (T. Lindgren); 0000-0001-9188-7425 (J. Pavlopoulos)



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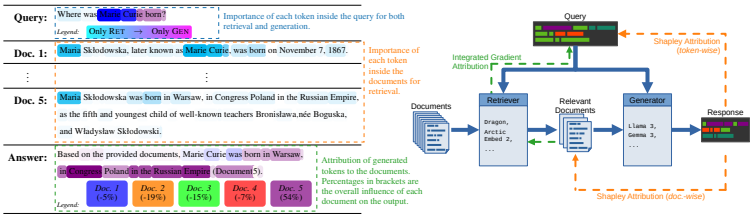


Figure 1: RAG-E overview (from [1]). (a) Example explanation obtained with Arctic Embed 2 and Llama 3: token saliency for the query and the documents on the RET side, token-wise and document-wise Shapley attribution on the GEN side. (b) Intrinsic IG explains the retriever; extrinsic Shapley attribution explains the generator.

how well the generator's use of the context follows the retriever's ranking, and show that it is better suited than rank or linear correlation and associated with downstream accuracy.

2. The RAG-E Framework

A RAG system consists of a retriever $\textsc{Ret}\colon (q, \mathcal{B}) \mapsto D$ that selects $D = \{d_1, \ldots, d_k\}$ from a knowledge base $\mathcal{B}$ given a query q, and a generator $y = \textsc{Gen}(q, D)$; q, d_i and y are token sequences. RAG-E tracks saliency through both components with *local* and *additive* methods, i.e., attributions sum to the difference between the explained and a baseline prediction, which makes retriever and generator explanations comparable in scale.

Retriever explanations: Integrated Gradients. Dense retrievers are differentiable, so an intrinsic method whose cost does not grow with the number of document tokens is available. We compute token-level attributions $\beta^{\text{ret}}_{i,k}$ of input token t_i towards the similarity score of document d_k by applying IG over the token embeddings, separately for the query and for each document. As IG baseline we replace every non-special token with the model's `[unk]` embedding at the same position, instead of the `[pad]` token suggested in the original IG paper; Section 3 shows that this choice is the most faithful one.

Generator explanations: stabilised Shapley values. Given the architectural variability of LLMs we explain GEN extrinsically, attributing each generated token t_j to each document d_k via Shapley Values, i.e., the marginal contribution of d_k across all subsets $D' \subseteq D$. Exact SV cost $O(2^{|D|})$ generator calls, so for larger $|D|$ we resort to KSHAP with $N = \min(2^{|D|}, 10 \cdot |D|)$ perturbed prompts. Two adaptations are essential for RAG: we do *not* apply KSHAP's marginal-contribution step, since retrieved documents often overlap and violate the independence assumption behind it; and, since GEN is autoregressive, we keep outputs comparable across perturbed sets D' by constrained generation, scoring each token of the answer generated for the full set D under the perturbed prompt [8, 9]. To reduce the approximation error towards the exact SV, PMCSHAP (Algorithm 1 in [1]) draws the N perturbations as *complementary pairs* (D'_1, D'_2) with $D'_1 \cap D'_2 = \emptyset$ and $D'_1 \cup D'_2 = D$ [16], then repeatedly fits KSHAP on M random subsets of size $N' < N$ of these pairs and averages the resulting attributions. The number of LLM calls stays fixed to N.

Quantifying RET–GEN agreement: WARG. Summing over tokens, $\bar{\beta}^{\text{ret}}_k = \sum_i \beta^{\text{ret}}_{i,k}$ recovers the retriever's similarity score for d_k, and $\bar{\beta}^{\text{gen}}_k = \sum_j \beta^{\text{gen}}_{k,j}$ is the influence of d_k on the generated answer. WARG first identifies the documents the generator actually relies on: with $K_+ = \{k \mid \bar{\beta}^{\text{gen}}_k > 0\}$, we normalise $\tilde{\beta}^{\text{gen}}_k = \bar{\beta}^{\text{gen}}_k / \sum_{k' \in K_+} \bar{\beta}^{\text{gen}}_{k'}$ and threshold at τ times the uniform-at-random level, $w_k = \mathbb{I}(\tilde{\beta}^{\text{gen}}_k > \tau/|D|)$, with $\tau = 1.5$ (justified empirically below). WARG is then the normalised difference in mean retriever importance between generation-relevant ($w_k = 1$) and irrelevant ($w_k = 0$) documents,

with $h = \sum_k w_k$ and the $\bar{\beta}_k^{\text{ret}}$ sorted in decreasing order:

$$WARG = \frac{\frac{1}{h}\sum_k \bar{\beta}_k^{\text{ret}} w_k - \frac{1}{|D|-h}\sum_k \bar{\beta}_k^{\text{ret}}(1-w_k)}{\frac{1}{h}\sum_{k=1}^{h} \bar{\beta}_k^{\text{ret}} - \frac{1}{|D|-h}\sum_{k=h+1}^{|D|} \bar{\beta}_k^{\text{ret}}}, \qquad WARG \in [-1, 1]. \tag{1}$$

The denominator is the best achievable separation given h: WARG is 1 when the generator uses exactly the retriever's top-h documents and negative when it prefers low-ranked ones.

3. Experimental Highlights

Setup. We audit six RAG pipelines: two dense retrievers, DRAGON [17] (110M parameters) and Snowflake Arctic Embed 2 [18] (568M), combined with three open-weight generators, Llama 3.1 8B [19], Qwen 2.5 7B [20] and Gemma 3 12B [21]. Design choices are validated on 200 queries from MS MARCO v2.1 [22] with five random passages each, measuring faithfulness as the Area Inside the Perturbation Curve (AIPC). The main analysis uses 200 queries from PopQA [23], 100 from QAMPARI [24] and 100 from TREC CAsT 2019 [25], with flat FAISS indexes. Documents are presented to Gen in four configurations: (C_1) the top-10 retrieved documents in rank order; (C_2) the top-5 followed by 5 random non-retrieved documents; (C_3) 5 random documents followed by the top-5; (C_4) the top-10 in shuffled order.

Design choices. For the retriever, the `[unk]` baseline yields the most faithful IG explanations for both models (AIPC 0.50 for DRAGON and 0.73 for Arctic Embed 2, against 0.41 and 0.67 with the `[pad]` baseline recommended in [13]): `[pad]` encodes empty space beyond the text boundary, whereas `[unk]` forces the model to infer meaning at that position. For the generator, PMCshap achieves the lowest mean squared error with respect to exact SV among Kshap, MCshap and their complementary-sampling variants, at a variance low enough for practical use. All Shapley-based methods and LIME reach comparable AIPC, roughly double that of ContextCite and MIRAGE.

What the explanations reveal. Nouns and proper nouns make up more than 70% (Arctic Embed 2) and about 50% (DRAGON) of the top-30% attributed retriever tokens, and on PopQA between 30% and 99% of the query words appear among the top-50% attributed document tokens: despite dense embeddings, retrieval still behaves much like keyword matching. On the generator side, document importance is approximately normally distributed around $\mu = \frac{1}{2|D|}$ with $\sigma = \frac{1}{|D|}$, and the few documents the answer depends on appear as positive outliers; this motivates the threshold $\tau = \mu + \sigma = 1.5$ in units of $\frac{1}{|D|}$; retriever and generator importances are otherwise not meaningfully correlated.

Table 1
Failure modes under C_1 for $k = 3$, in % of queries. *Wasted Retrieval*: the top-ranked document receives Gen rank $> k$. *Noise Distraction*: the most influential document for Gen has Ret rank $> k$.

		Wasted Retrieval [%]			Noise Distraction [%]		
Ret	Gen	PopQA	TREC CAsT	QAMPARI	PopQA	TREC CAsT	QAMPARI
DRAGON	Llama	77.5	40.0	23.0	76.0	36.0	22.0
	Qwen	72.0	32.0	16.0	79.0	36.0	15.0
	Gemma	40.5	47.0	41.0	38.5	48.0	38.0
Arctic Embed 2	Llama	76.5	34.0	28.0	83.0	30.0	29.0
	Qwen	74.5	35.0	21.0	79.5	33.0	23.0
	Gemma	41.5	45.0	38.0	31.5	40.0	44.0

Table 2
Mean Spearman r, Pearson r and WARG under C_2 (higher is better), C_3 (lower is better) and C_4 (closer to 0 is better), averaged over the six RET–GEN pairs (per-pair values in [1]).

	C_2 (top-k first)			C_3 (random first)			C_4 (shuffled)		
Dataset	Sprmn.	Prsn.	WARG	Sprmn.	Prsn.	WARG	Sprmn.	Prsn.	WARG
PopQA	0.44	0.49	**0.68**	−0.13	−0.17	**−0.28**	0.09	0.06	**0.04**
TREC CAsT	0.42	0.43	**0.52**	−0.12	−0.14	**−0.19**	**0.14**	0.15	0.17
QAMPARI	0.48	0.51	**0.63**	−0.27	−0.28	**−0.33**	0.07	0.07	0.07

Failure modes. Table 1 quantifies two consequences of this mismatch: *Wasted Retrieval*, when the retriever's top document is largely ignored by the generator, and *Noise Distraction*, when the generator relies most on a document ranked outside the top-k. For $k = 3$, Wasted Retrieval ranges from 16% (DRAGON + Qwen on QAMPARI) to 77.5% (DRAGON + Llama on PopQA) and Noise Distraction from 15% to 83% (Arctic Embed 2 + Llama on PopQA): the high end and the strong dataset dependence indicate a systematic disagreement.

Validating WARG. To answer our RQ we compare WARG with Spearman and Pearson correlation between retriever and generator importances under controlled prompt orderings, assuming for C_2–C_4 that retriever importance decreases linearly with rank (Table 2). When the retrieved documents come first (C_2) WARG is distinctly higher than both correlations; when they come last (C_3) it is consistently lower; when the order is shuffled (C_4) all metrics are close to zero. WARG thus separates agreement from disagreement more sharply: it models the generator's focus on a few salient documents, whereas Spearman is sensitive to noise in the tail of near-zero importances and Pearson assumes a linearity that near-binary importances violate.

Finally, WARG is connected to downstream quality: on PopQA and QAMPARI, an answer containing a ground-truth label is more than twice (PopQA) and more than four times (QAMPARI) as likely to have $WARG > 0.5$ than an answer that does not; a χ^2 test of independence confirms the association (PopQA: $\chi^2 = 16.5$, $p < 10^{-4}$; QAMPARI: $\chi^2 = 35.2$, $p < 10^{-8}$). High WARG does not guarantee a correct answer, but when the two components agree the relevant documents are available and used.

4. Discussion and Outlook

RAG-E turns the RET–GEN interaction into a measurable quantity, and WARG is, to our knowledge, the first principled metric for it. For IR, three observations stand out. First, dense retrievers still look like keyword matchers over content-bearing nouns, and a ranked list alone cannot tell whether a downstream generator will use its documents. Second, the tested generators avoid primacy bias in the short contexts used here (50–90 words per document), attributing by content rather than position when the order is randomised, unlike the lost-in-the-middle effect [26] on longer inputs. Third, misalignment is frequent enough that it should be monitored: WARG can audit deployed pipelines, compare retriever–generator pairings and, in future work, serve as an optimisation signal for query rewriting, reranking or RL-based RAG. Limitations include the cost of Shapley-based attribution, which still scales with $|D|$; the restriction to dense retrievers and autoregressive generators; and the short documents used here.

Declaration on Generative AI

During the preparation of this work, the authors used Claude (Anthropic) in order to: draft and condense text, grammar and spelling check. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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